# Evolution of Universe



Tue 11/25

## Basic introduction to geography -

Systematic Study of Earth

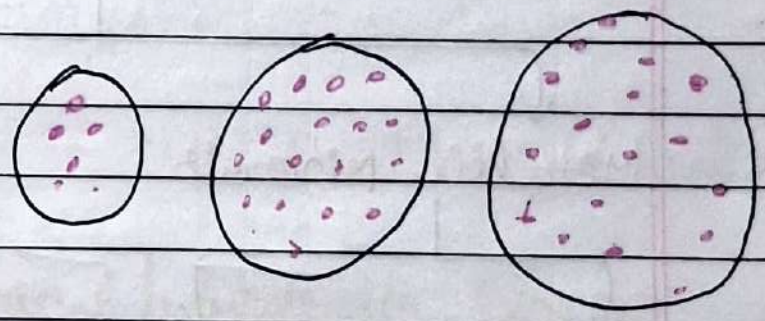
"Geo" → Earth, "Graphein" → To write

## Origin of Universe -

### 1. The Steady State Theory -

Number of galaxies in the observable universe remains constant.

Matter is constantly created as universe expands



### 2. Big bang theory -

Inflation → First → First Nuclei
(Initial expansion) particles
→ First light → First Stars → Galaxies
and dark matter

07/02/26

Minerals – Naturally occurring combinations of elements with specific physical and chemical properties.

- Inorganic substances or organic (fossil fuels).
- They are found in earth crust as magma is one of the primary source of minerals.
- Composed of one or more elements and occur in various forms, colours or hardness levels.

Minerals

- Metallic Minerals
  - Ferrous: Ex-Iron, Manganese, Nickel etc. Use – Steel production, construction, tools & machinery
  - Non-Ferrous: Copper, Tin, lead, zinc, Bauxite, Natural gas, Gold, Silver. Use – Electronics, power generation & as a fuel source, aerospace, jewellery & renewable energy
  - Fuel / Energy: Coal, petrol, Natural gas, Uranium, Thorium. Use – power generation & as a fuel source
  - Industrial: Limestone, Dolomite, Gypsum, Quartz. Use – Cement, glass, fertilizers, plaster
- Non-Metallic Minerals
  - Gemstones: Diamond, Emerald, Ruby. Use – jewellery and decorative purposes

Flowchart – Type of Minerals



1. Metallic → Contain metal elements in chemical composition, always inorganic (geological processes).
   - & serve as sources of metals.
   - (a) Ferrous – Contain iron and are magnetic.
   - (b) Non-ferrous – do not contain iron and are non-magnetic.
2. Non-metallic → do not contain metals.
   - & either organic or inorganic.

Major sites in India – Metallic | World Country

1. Iron – Odisha, Jharkhand | Australia, Brazil
2. Bauxite – Odisha, Gujarat | Australia, China
3. Gold – Karnataka | China, Russia
4. Zinc & lead – Rajasthan | China, Peru, Australia

Non-Metallic –

1. Limestone – Madhya pradesh, Rajasthan | China
2. Coal – Jharkhand, Odisha | China, Indonesia
3. Mica – Andhra pradesh | Madagascar
4. Salt – Gujarat | China, US

Energy resources – Sources of energy used to generate heat, light & power.

## Sources of Energy

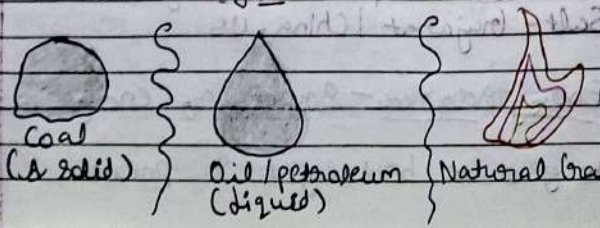
- Main Sources of energy
  - Conventional Sources
    - Commercial
      - Coal
      - Oil & Natural gas
      - Thermal power
    - Non-Commercial
      - Firewood (fuel wood)
      - Hydro power
      - Nuclear power
- Non-Conventional Sources
  - Solar energy
  - Wind
  - Tidal
  - Geothermal
  - Biomass based

### Fossil fuels (Time + Pressure + Heat)

Formed from the remains of ancient plants and animals, buried deep inside the earth from millions of years.

* The immense heat and pressure from the earth transformed them into concentrated energy sources.

### Types



Coal (A solid) | Oil / petroleum (liquid) | Natural Gas (Gas)



1. **Coal** - Also known as black diamond and buried sunshine

Formation -

[Ancient plants] + [Time + pressure + Heat] → C (coal)

Types - Anthracite > Bituminous > Lignite
High Moderate Low
Based on concentration of Carbon

Uses - Steam, steel manufacturing, cook fuel

2. **Petroleum / Oil** - known as liquid gold and Black gold because of great commercial importance.

Formation -

[Sea animals & plants] + [Time + pressure + Heat] → P (Petroleum) + G (Natural Gas)

Uses - Petrol fuel, water repellent, fibres

3. **Natural Gas** - Clean & Non toxic

Formation -

[Phyto plankton] / [Zoo] + [Organic material] + [pressure + Heat] → Kerogen (without oxygen)

[Kerogen] + [Temp 160-180°C] → (Natural Gas)

Uses – Generating power, automobile fuels.

Fossil fuel formation flowcharts –

1. Coal – Dense forests and swamy areas
↓
Dead plants fall into swamps
↓
Buried under water & Mud
↓
X No oxygen (Anaerobic conditions)
↓
8 Million of years (High pressure + High temp)
↓
Peat (dense, spongy, soil like material)
↓
Lignite (High oxygen & hydrogen, low carbon)
↓
Bituminous Coal (Moderate)
↓
Anthracite Coal (Best Quality) (High carbon content)

Coal mining is done through subterranean operations.

2. Formation of petroleum (Crude oil) & Natural Gas –

This is formed from warm, shallowed seas.



Shallow seas & Ocean basins
↓
Dead Marine Microorganisms (Phytoplankton + Zooplankton)
↓
Decomposition on sea floor
↓
Burial under sediments (Sand, Clay, Silt)
↓
X Absence of oxygen (Anaerobic conditions)
↓
Anaerobic bacterial action
↓
Formation of kerogen (Complex organic matter)
↓

| | |
|---|---|
| Temperature 60°C – 120°C + High pressure (Oil window) | Temperature 120°C – 200°C + Very high pressure (Gas window) |
| ↓ | ↓ |
| Petroleum (Crude oil) | Natural Gas (Methane rich) |
| ↓ | ↓ |
| Migration & Tapping (Porous rocks + Cap Rocks) | Accumulates above petroleum (lightest fossil fuel) |
| ↓ | ↓ |
| Petrol, Diesel, Kerosene, LPG, Lubricating oil, Bitumen/Asphalt | Compressed Natural Gas (CNG), Piped Natural Gas (PNG) |

Sequence of fossil fuels (Top to Bottom in Earth) -

Natural Gas
(lightest - floats at the top)
↓
Petroleum (Crude oil)
(medium density)
↓
Coal
(heaviest - found deepest)

Top ↓ Bottom

Continents & Countries by fossil fuel reserves -

Coal - list in 2018, mid-million

| | Country | Anthracite & bituminous | | Subbituminous & lignite | | Total | |
|---|---|---|---|---|---|---|---|
| | | Tonnes (mil) | % | Tonnes (mil) | % | Tonnes (mil) | % |
| 1. | United States | 220,167 | 30% | 30,052 | 9.4% | 250,219 | 24% |
| 2. | Russia | 69,634 | 9.5 | 90,370 | 28.4 | 160,364 | 15 |
| 3. | Australia | 70,927 | 9.7 | 76,508 | 23.9 | 147,435 | 14 |
| 4. | China | 130,851 | 17.8 | 7,968 | 2.5 | 138, | 13 |



| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | 819 | |
| 5. | India | 96,468 | 13.1 | 4,895 | 1.5 | 101,363 | 10 |

Petroleum -

Organization of Petroleum Exporting Countries (OPEC) -

Permanent, Intergovernmental organization, created at Baghdad Conference in Sep 1960 by Iran, Iraq, Kuwait, Saudi Arabia and Venezuela. Currently it has 13 members.

Proven reserve (millions of barrel)

| Country | EIA (Energy Information administration) | OPEC | BP (British petroleum) |
|---|---|---|---|
| Venezuela | 304,000 | 303,220 | 303,800 |
| Saudi Arabia | 267,000 | 267,230 | 297,600 |
| Iran | 209,000 | 208,600 | 155,600 |
| Iraq | 201,000 | 145,020 | 175,000 |
| Canada | 170,000 | 4,731 | 172,200 |

- Most oil rich continent : Asia
- OPEC countries hold : ~79% of global oil reserves
- Venezuela oil : 17% of global oil, largest hold by single country

Middle east - Region around centered around western asia & Egypt connecting Asia, Africa & Europe, and holding major historical, cultural, religious and energy (oil & gas) significance

| Region | Countries | Capitals |
|---|---|---|
| 1. Western Asia (core middle east) | Saudi Arabia, UAE, Qatar, Kuwait, Oman, Yemen, Iraq, Iran, Israel, Jordan, Lebanon, Syria, Bahrain | Riyadh, Abu dhabi, Doha, Kuwait city, Muscat, Sana'a, Baghdad, Tehran, Jerusalem, Amman, Beirut, Damascus, Manama |
| 2. Transitional | Turkey | Ankara |
| 3. North Africa | Egypt | Cairo |

Total (most accepted) : 15 countries

The Red sea, Persian gulf, and Strait of Hormuz are vital for international trade routes and



the supply of energy.

Reason of oil in middle east -

The Tethys sea was a vast, warm, shallow sea that existed between the ancient continents of Gondwana and Laurasia.

- During the Mesozoic era, much of today's middle east lay beneath this sea.

Oil formation -

- Tethys was rich in plankton, algae and marine organisms.
- When this organisms died, they form organic rich sediment which became the primary source rocks for petroleum.

Natural Gas -

Proven reserves ($km^3$)

| Country | U.S EIA (2021) | OPEC (2018) | BP (2020) |
|---|---|---|---|
| Russia | 47,800 | 47,760 | 37,400 |
| Iran | 34,400 | 33,988 | 32,100 |
| Qatar | 23,900 | 23,861 | 24,700 |
| U.S | 17,710 | 9,067 | 12,600 |